The Entropy Cost of Vanishing Activity

A Trichotomy for Exact Common Causal Seeds

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Abstract

We examine a time-inhomogeneous binary controlled process simulated by a single random seed that is exact for *every* adaptive policy and assigns a consistent response at shared histories. Suppose every conditional probability of output 1 at time t is at most $p_t \in (0, 1/2]$, and this upper bound is attainable at every history. If C_n denotes the minimum Shannon entropy of an exact common causal seed through time n, then

$$\sum_{t=1}^{n} h_2(p_t) \le C_n \le \sum_{t=1}^{n} h_2(p_t) + 2\sum_{t=1}^{n} p_t.$$

A direct construction yields a sharp qualitative trichotomy for countably supported seeds on the infinite horizon. In particular, for $p_t = 1/[t(\ln t)^2]$, $t \ge 3$, one common countable seed makes every policy permanently inactive after a finite random time almost surely, yet $C_n = \log_2 \ln n + O(1)$. Thus almost-sure cessation of activity does not imply a finite Shannon entropy cost. The claims are proved here directly; no claim of priority or machine-checked formalization of this note is made.

1 Model and notion of exactness

At each time $t \ge 1$ and every finite output–action history h available before that time, let $\mathcal{A}_t(h)$ be a nonempty finite action set. An action $a \in \mathcal{A}_t(h)$ induces a binary output Y_t with prescribed conditional law

$$\mathbb{P}(Y_t = 1 \mid h, a) = q_t(h, a), \qquad 0 \le q_t(h, a) \le p_t \le \tfrac{1}{2}, \qquad p_t > 0. \tag{2.1}$$

Assume throughout that *for every* t, h, there is an available action $a_t^\star(h)$ with $q_t(h, a_t^\star(h)) = p_t$. Actions may depend on the entire past, and the values $q_t(h, a)$ need not be stationary.

An *exact common causal seed* is a random variable S together with deterministic functions

$$F_t(s, h, a) \in \{0, 1\} \qquad (t \ge 1),$$

such that, for every deterministic adaptive policy π, the transcript generated recursively by $Y_t = F_t(S, h_{t-1}, a_t)$ has the law specified by the kernels q_t. In particular, the same seed value fixes the response at a given history and action irrespective of which policy reaches that history. A seed is *countably supported* if its law is supported on a countable set. For such a seed its Shannon entropy (in bits) is

$$\mathrm{H}(S) = -\sum_{s} \mathbb{P}(S = s) \log_2 \mathbb{P}(S = s) \in [0, \infty].$$

Write C_n for the infimum of $\mathrm{H}(S)$ over countably supported exact common causal seeds through horizon n. Since there are finitely many histories and response strategies to a fixed finite horizon, the same infimum can be taken over finite seeds. Set

$$h_2(u) = -u \log_2 u - (1 - u) \log_2(1 - u), \quad h_2(0) = h_2(1) = 0.$$

2 A finite-horizon entropy sandwich

Theorem 1 (Sparse-activity bound)**.** *For every integer $n \geq 1$,*

$$\boxed{\sum_{t=1}^{n} h_2(p_t) \;\leq\; C_n \;\leq\; \sum_{t=1}^{n} h_2(p_t) + 2\sum_{t=1}^{n} p_t.} \tag{1}$$

These bounds are independent of the number of policies and of the history-dependence of the kernels.

Proof. Lower bound. Choose the deterministic policy $\pi^\star$ that selects an action $a_t^\star(h)$ with success probability p_t at every history. Its conditional output law at time t does not depend on the past. Therefore its output vector has the product law

$$(Y_1, \ldots, Y_n) \sim \bigotimes_{t=1}^{n} \text{Bernoulli}(p_t), \qquad \mathrm{H}(Y_1, \ldots, Y_n) = \sum_{t=1}^{n} h_2(p_t).$$

The output vector is a deterministic function of any exact seed under this fixed policy, so $\mathrm{H}(S) \geq \mathrm{H}(Y_1, \ldots, Y_n)$. Taking the infimum proves the first inequality.

Upper bound. Independently at each time t, sample $Z_t \in \{0, 1, 2, \ldots\}$ according to

$$\mathbb{P}(Z_t = 0) = 1 - p_t, \qquad \mathbb{P}(Z_t = k) = p_t 2^{-k} \quad (k \geq 1). \tag{2}$$

For every (t, h, a), select binary digits $b_k(t, h, a) \in \{0, 1\}$ satisfying

$$\frac{q_t(h, a)}{p_t} = \sum_{k \geq 1} b_k(t, h, a) 2^{-k}. \tag{3}$$

Either binary expansion may be used at a dyadic rational; use the all-ones expansion for the endpoint 1. Define the total response strategy

$$F_t((z_j)_{j \geq 1}, h, a) = \begin{cases} 0, & z_t = 0, \\ b_{z_t}(t, h, a), & z_t \geq 1. \end{cases}$$

For a fixed history h and action a, its response probability is exactly

$$\mathbb{P}(F_t(S, h, a) = 1) = p_t \sum_{k \geq 1} 2^{-k} b_k(t, h, a) = q_t(h, a).$$

Furthermore, under any policy, the history and the next selected action are functions of $Z_1, \ldots, Z_{t-1}$. Since Z_t is independent of those coordinates, the same equality holds *conditional on any reachable history.* Consequently the entire transcript law is exact for every adaptive policy, using one shared response strategy.

Let G be a positive-integer-valued random variable with $\mathbb{P}(G = k) = 2^{-k}$, $k \geq 1$. Then $\mathrm{H}(G) = \sum_{k \geq 1} k 2^{-k} = 2$ bits. Splitting Z_t according to the event $Z_t \neq 0$ gives

$$\mathrm{H}(Z_t) = h_2(p_t) + p_t \mathrm{H}(G) = h_2(p_t) + 2p_t. \tag{4}$$

The first n coordinates form a countable exact seed, with entropy

$$\mathrm{H}(Z_1, \ldots, Z_n) = \sum_{t=1}^{n} \mathrm{H}(Z_t).$$

To obtain a *finite* seed, map each vector $(z_1, \ldots, z_n)$ to its complete deterministic response table for all histories and actions up to time n. There are only finitely many such tables; the map preserves exactness and cannot increase Shannon entropy. Equations (4) and (1) follow. □

3 Infinite-horizon trichotomy

Theorem 2 (Countable-seed classification)**.** *Under the same assumptions, exactly one of the following occurs:*

(i) *If* $\sum_{t\geq 1} p_t = \infty$*, no countably supported exact common causal seed exists on the infinite horizon.*

(ii) *If* $\sum_{t\geq 1} p_t < \infty$ *but* $\sum_{t\geq 1} h_2(p_t) = \infty$*, a countably supported exact common causal seed exists, but every such seed has infinite Shannon entropy.*

(iii) *If* $\sum_{t\geq 1} h_2(p_t) < \infty$*, there exists a countably supported exact common causal seed of finite entropy.*

In cases (ii) *and* (iii)*, a seed can be chosen for which, almost surely, there is one finite time after which* all *adaptive policies simultaneously produce only zeros.*

Proof. For (i), use the maximizing policy $\pi^\star$ from the preceding proof. For any particular infinite output sequence $y = (y_t)_{t\geq 1}$,

$$\mathbb{P}_{\pi^\star}((Y_t)_{t\geq 1} = y) = \prod_{t\geq 1} p_t^{y_t}(1-p_t)^{1-y_t} \leq \prod_{t\geq 1}(1-p_t) = 0.$$

The last equality follows from $\sum_t p_t = \infty$ and $\ln(1-p_t) \leq -p_t$. Thus the transcript law of $\pi^\star$ is nonatomic. The deterministic image of a countably supported seed is supported on a countable set, and hence has an atom. This contradiction proves (i).

Suppose $\sum_t p_t < \infty$ and use the independent variables in (2) for *all* t. By the first Borel–Cantelli lemma,

$$\mathbb{P}(Z_t \neq 0 \text{ for infinitely many } t) = 0.$$

The set of sequences $(z_t)_{t\geq 1}$ with only finitely many nonzero coordinates is countable, being a countable union of finite products of countable sets. Hence this construction provides a countably supported exact seed $S = (Z_t)_{t\geq 1}$. With

$$T(S) = \sup\{t : Z_t \neq 0\}, \qquad \sup \varnothing = 0,$$

we have $T(S) < \infty$ almost surely, and $F_t(S, h, a) = 0$ for every $t > T(S)$ and every history–action pair (h, a). This establishes the simultaneous-cessation statement.

In case (ii), the maximizing-policy transcript has entropy $\sum_{t=1}^n h_2(p_t)$ through time n. Since every such finite transcript is a deterministic image of any exact infinite-horizon seed S,

$$\mathrm{H}(S) \geq \sup_n \sum_{t=1}^n h_2(p_t) = \infty.$$

In case (iii), the inequality $h_2(p_t) \geq p_t$ for $0 < p_t \leq \frac{1}{2}$ implies $\sum_t p_t < \infty$, and the constructed seed satisfies

$$\mathrm{H}(S) = \sum_{t\geq 1} \mathrm{H}(Z_t) = \sum_{t\geq 1} [h_2(p_t) + 2p_t] < \infty.$$

For completeness, the entropy identity follows from independence and the monotone limit of the information of finite prefixes; since the infinite product is supported on finite-activity sequences, the limit is $-\log_2 \mathbb{P}(S)$ almost surely. This proves (iii). □

4 A doubly logarithmic information cost

Corollary 3 (Almost-sure silence with unbounded entropy)**.** *Start at $t = 3$ and take*

$$p_t = \frac{1}{t(\ln t)^2}, \qquad t \geq 3.$$

For every controlled binary process satisfying the hypotheses with this choice of p_t,

$$\boxed{C_n = \log_2 \ln n + O(1) \qquad (n \to \infty).} \tag{5}$$

There exists a single countably supported exact causal seed for all horizons, with an almost-surely finite last active time for every policy, but every such seed has infinite Shannon entropy.

Proof. The integral test gives $\sum_{t\geq 3} p_t < \infty$. As $u \downarrow 0$,

$$h_2(u) = \frac{u \ln(1/u) + u + O(u^2)}{\ln 2}.$$

Using $\ln(1/p_t) = \ln t + 2 \ln \ln t$, we obtain

$$h_2(p_t) = \frac{1}{\ln 2} \frac{1}{t \ln t} + O\left(\frac{1 + \ln \ln t}{t(\ln t)^2} \right).$$

The error term is summable, while $\sum_{t=3}^{n} (t \ln t)^{-1} = \ln \ln n + O(1)$. Therefore

$$\sum_{t=3}^{n} h_2(p_t) = \log_2 \ln n + O(1).$$

The term $2 \sum_{t=3}^{n} p_t$ in (1) is uniformly bounded, proving (5). The remaining statements follow from Theorem 2. □

Remark 4 (Number of events versus time of the last event)**.** *For the explicit seed above let $N = \sum_{t\geq 3} \mathbf{1}_{\{Z_t \neq 0\}}$, so $\mathbb{E}N = \sum_{t\geq 3} p_t < \infty$. In contrast, the last potential activity time $T = \sup\{t \geq 3 : Z_t \neq 0\}$ satisfies*

$$\mathbb{P}(T \geq m) = 1 - \prod_{t\geq m} (1 - p_t) \sim \sum_{t\geq m} p_t \sim \frac{1}{\ln m} \qquad (m \to \infty),$$

and therefore $\mathbb{E}T = \infty$. Thus finite expected activity count*, almost-surely finite* cessation time*, and finite expected* cessation time *are distinct properties.*

5 Interpretation and scope

This example isolates a precise informational phenomenon: *a realization can contain only finitely many active times while the information needed to specify which times are active has infinite mean.* The divergence is slow but quantitative: the optimal finite-horizon seed entropy grows with leading coefficient one in $\log_2 \ln n$.

The result concerns a deliberately nonstationary family of kernels. It does not establish any stationary logarithmic excess theorem, identify a physical mechanism for the passage of time, or make a claim about consciousness. It is compatible with the common-causal-seed framework studied in *One Seed for Every Adaptive Policy*:

`https://github.com/jendamikulik/finite-Shannon-theorem`

The direct arguments above are self-contained; their novelty relative to the broader literature has not been assessed, and this particular trichotomy has not been claimed here as formally verified in Lean.